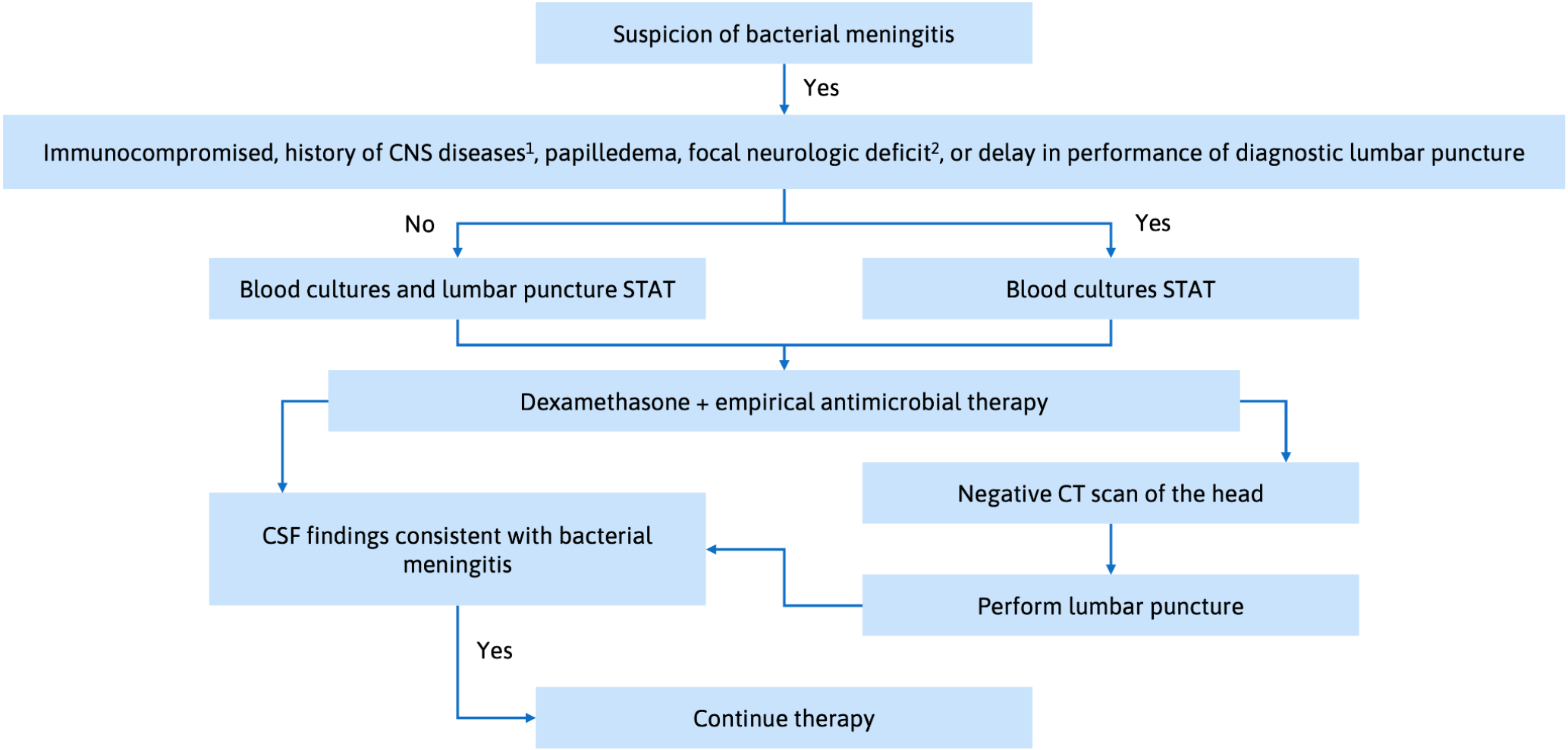
Central Nervous System Infections

Management Algorithm of Bacterial Meningitis



1. Include those associated with CSF shunts, hydrocephalus, or trauma, those occurring after neurosurgery, or various space-occupying lesions.
2. Palsy of cranial nerve VI and VII is not an indication to delay lumbar puncture.

Vaccinations

1	For all patients with confirmed pneumococcal and meningococcal meningitis, consider vaccinations post-recovery.
2	For all patients with proven or suspected CSF leak, consider pneumococcal vaccinations with PCV13. For further information regarding pneumococcal vaccination schedule please visit: **https://www.cdc.gov/vaccines/vpd/pneumo/downloads/pneumo-vaccine-timing.pdf**.

CHAPTER 3.3	**CENTRAL NERVOUS SYSTEM INFECTIONS**		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Meningitis			
<50 years old Immunocompetent *S pneumoniae* *N meningitidis* *H influenzae*	Ceftriaxone 2 g IV q12H		• Duration: 10 - 14 days • Take blood C&S and perform CSF examination as soon as possible. • Perform CT brain before LP for patients with: ◦ History of CNS diseases (mass lesion, CVA) ◦ New onset seizure ◦ Papilledema ◦ Altered consciousness ◦ Focal neurologic deficit • Take cultures before empiric antibiotic therapy (at least blood C&S if LP must be delayed). • Do not wait for a CT scan or an LP to start therapy.
>50 years old *S pneumoniae* Listeria *N meningitidis* *H influenzae* Group B Streptococcus	Ceftriaxone 2 g IV q12H + Ampicillin 2 g IV q4H		
Immunocompromised (including alcoholism,	Ceftriaxone 2 g IV q12H +		

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
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pregnant, debilitated) *S pneumoniae* Listeria Gram-negative bacteria	Ampicillin 2 g IV q4H		• IDSA (2004) suggests LP avoidance for patients with these conditions: • HIV/AIDS • Receiving immunosuppressive therapy • Post-transplant
Penetrating Head Injury *S epidermidis* *S aureus* *P acnes* *S pneumoniae* Facultative and aerobic gram-negatives including: *P aeruginosa* *A baumannii* (possible MDR)	Vancomycin 20 mg/kg IV single dose (max 2 g), then 15 mg/kg (max 1 g) IV q8-12H + Ceftazidime 2 g IV q8H	Vancomycin 20 mg/kg IV single dose (max 2 g), then 15 mg/kg (max 1 g) IV q8-12H + Cefepime 2 g IV q8H	• Take cultures before empiric antibiotic therapy (at least blood C&S). • **Duration is typically 10 - 14 days depending on organism:** • *Streptococcus pneumoniae*: 10-14 days • *Neisseria meningitidis*: 7 days • *Haemophilus influenzae*: 7 days • Listeria: 21 days • Gram-negative bacilli: 21 days • Vancomycin: Aim for serum trough level of 15-20 mg/L. • **Patients who have experienced anaphylaxis to beta-lactams, please refer ID.**

CHAPTER 3.3	CENTRAL NERVOUS SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Trauma With Basilar Skull Fracture *S pneumoniae* *H influenza* *S pyogenes*	**Antibiotics are not required unless infected.** **If evidence of infection:** Vancomycin 20 mg/kg IV single dose (max 2 g), then 15 mg/kg (max 1 g) IV q8-12H + Ceftazidime 2 g IV q8H		
Ventriculitis			
Healthcare-associated Ventriculitis, including post-neurosurgery ventriculitis	Vancomycin 20 mg/kg IV single dose (max 2 g), then 15 mg/kg (max 1 g) IV q8-12H + Ceftazidime 2 g IV q8H	Vancomycin 20 mg/kg IV single dose (max 2 g), then 15 mg/kg (max 1 g) IV q8-12H + Cefepime 2 g IV q8H	• Ensure that CSF and blood cultures are sent. • If initial cultures are negative, and suspicion is high, to contact the lab to incubate for up to 10 days. • If a shunt is present, and infected, removal is essential. • **When to re-implant shunt:** • CoNS/*P acnes* with no CSF abnormality and negative cultures after 48 hours of externalisation: third day after

<table>
<tr><th>CHAPTER 3.3</th><th colspan="3">CENTRAL NERVOUS SYSTEM INFECTIONS</th></tr>
<tr><th rowspan="2">Infections and Likely Organisms</th><th colspan="2">Suggested Treatment</th><th rowspan="2">Notes</th></tr>
<tr><th>Preferred</th><th>Alternative</th></tr>
<tr><td></td><td></td><td></td><td>removal.
• CoNS/P acnes with CSF abnormality and repeat negative cultures: after at least 7 days of antimicrobial therapy.
• If repeat culture is positive, continue antibiotics: re-implant after 7-10 days of negative culture.
• S aureus or gram-negative bacilli: 10 days after CSF culture negative.
• Patients who have experienced anaphylaxis to beta-lactams, please refer ID.</td></tr>
<tr><td colspan="4">Brain Abscess</td></tr>
<tr><td>Primary
<u>Unknown source</u>
S aureus
Streptococci
Gram-negative
Anaerobes

Sinusitis
Streptococci (including S pneumoniae)
Anaerobes</td><td>Ceftriaxone 2 g IV q12H
+
Metronidazole 500 mg IV q8H</td><td>Penicillin Allergy:
Please refer to ID team and may start:
Ciprofloxacin 400 mg IV q8H
+
Metronidazole 500 mg IV q8H</td><td>• Duration: Treat until response is seen by neuroimaging.
• Surgical emergency is a must for drainage of the abscess.
• For primary brain abscess with unknown source, add Vancomycin if S aureus is suspected.</td></tr>
</table>

CHAPTER 3.3	CENTRAL NERVOUS SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Chronic Otitis Media Gram-negative Streptococci Anaerobes	Cefepime 2 g IV q8H + Metronidazole 500 mg IV q8H		• Duration: Treat until response is seen by neuroimaging. • Surgical emergency is a must for drainage of the abscess.
Post-surgical, Traumatic or Subdural Empyema Staphylococci Enterobacteriaceae	Vancomycin 20 mg/kg IV single dose (max 2 g), then 15 mg/kg (max 1 g) IV q8-12H + Ceftazidime 2 g IV q8H	Vancomycin 20 mg/kg IV single dose (max 2 g), then 15 mg/kg (max 1 g) IV q8-12H + Cefepime 2 g IV q8H	
Cyanotic Heart Disease Streptococci (*S viridans*)	Ceftriaxone 2 g IV q12H		

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Immunocompromised Nocardia *Toxoplasma gondii* Cryptococcus Aspergillus Scedosporium	Please refer to the ID team.		
Encephalitis			
Herpes (HSV) Rarely VZV, EBV, HHV-6 Other rare treatable causes: *M tuberculosis,* Listeria, cat-scratch disease, Mycoplasma	Acyclovir 10 mg/kg IV q8H		• Duration: 14 days. • Add Doxycycline 100 mg PO q12H, if suspect Rickettsia, Mycoplasma.

References:

1. Infectious Diseases Society of America (IDSA): Clinical practice guidelines for healthcare-associated ventriculitis and meningitis (2017)
2. IDSA: Practice guidelines for the management of bacterial meningitis (ARCHIVED; 2004)
3. European Society of Clinical Microbiology and Infectious Diseases (ESCMID): Guideline for the diagnosis and treatment of acute bacterial meningitis (2016)
4. UpToDate: Initial therapy and prognosis of community-acquired bacterial meningitis in adults (Accessed July 2024).
5. UpToDate: Healthcare-associated meningitis and ventriculitis in adults: Treatment and prognosis (Accessed July 2024).